Expense Tracker with Smart Insighter

Development Planning Document

Project Schedule, Tools & Team Roles

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MP Online Internship Project

1. Project Overview:

The Development Planning Document (DPD) outlines the structured approach followed during the development of the **Expense Tracker with Smart Insighter**. It defines the development strategy, phases, tools, timelines, and methodologies used to ensure systematic project execution. This document reflects an industry-oriented workflow, focusing on planning, modular development, testing, and iterative improvement.

2. Development Phases:

Phase	Phase Name	Tasks	Duration	Status
Phase 1	Planning & Analysis	Requirement gathering, defining objectives, BRD preparation, system design	Week 1	Completed
Phase 2	Backend Setup	FastAPI project setup, database models creation, user authentication system	Week 2	Completed
Phase 3	Core API Development	Expense CRUD APIs, budget management APIs, data validation logic	Week 3	Completed
Phase 4	Frontend Development	React UI development, routing setup, dashboard design, Axios integration	Week 4	Completed
Phase 5	Intelligence Module	Expense categorization logic, prediction feature, anomaly detection	Week 5	Completed
Phase 6	Testing & Debugging	Unit testing, API testing, bug fixing, validation checks	Week 6	Completed
Phase 7	Final Review	Documentation, UI refinement, code cleanup, final system validation	Week 7	Completed

3. Tools and Technologies Used:

Layer	Technology	Version	Purpose
Backend Framework	FastAPI	0.115+	REST API development

Server	Uvicorn	0.30+	ASGI server for running FastAPI
Database ORM	SQLAlchemy	2.0+	Database abstraction and ORM
Database	PostgreSQL	14+	Persistent data storage
Auth Library	python-jose	3.3.0	JWT token generation and validation
Password Hashing	Passlib + bcrypt	1.7.4 / 4.x	Secure password hashing
Data Validation	Pydantic v2	2.x	Data validation and serialization
Frontend Framework	React	18+	UI component-based development
Build Tool	Vite	5+	Fast frontend build tool and dev server
Styling	Tailwind CSS	3.x	Utility-first CSS framework
HTTP Client	Axios	1.x	API communication from frontend
Database Driver	psycopg2	2.9+	PostgreSQL adapter for Python

4. Team Roles & Responsibilities:

Role	Assigned To	Responsibilities
Full Stack Developer	SELF	End-to-end development of backend APIs and frontend UI
Backend Developer	SELF	FastAPI setup, DB models, JWT auth, all routers and services
Frontend Developer	SELF	React pages, routing, Axios integration, UI components
Database Designer	SELF	MySQL schema design, SQLAlchemy model definition
Tester	SELF	Swagger UI testing, manual test cases, bug verification
Documentation Lead	SELF	BRD, DDD, RTM, Test Docs preparation

5. Development Tools:

Tool	Category	Usage
Visual Studio Code (VS Code)	IDE	Primary code editor for frontend and backend development
Postman / Swagger UI	API Testing	Testing and validating REST API endpoints
pgAdmin / DBeaver	Database Tool	PostgreSQL database management, schema design, and query execution
Git + GitHub	Version Control	Source code management and version tracking
npm / pip	Package Manager	Managing frontend and backend dependencies
Vite Dev Server	Frontend Development	Development server with hot reload for React UI

6. Task Management Approach:

The development tasks were managed in a structured manner by dividing the project into smaller modules. Each module was completed sequentially, ensuring clarity and reducing complexity. Basic task tracking was maintained to monitor progress and ensure timely completion.

7. Risk Management:

Potential risks identified during development included:

- Integration issues between frontend and backend
- Incorrect API responses
- Data inconsistency
- UI bugs

Mitigation strategies:

- Regular testing after each module
- Debugging using logs
- Step-by-step integration

8. Version Control Strategy:

The project uses Git for version control. Code changes were tracked regularly, and updates were pushed to GitHub. This ensured safe code management, rollback capability, and proper version tracking.

9: Conclusion:

The Development Planning Document reflects a structured and systematic approach to building the Expense Tracker with Smart Insighter. By following a phased development strategy and integrating testing at each stage, the project ensures reliability, scalability, and maintainability. The planning approach aligns with industry practices and provides a strong foundation for future enhancements.